

Column Space

Definition

Let A be an $m \times n$ matrix.

- The **column space** of A is the span of the columns of A and is denoted by $\text{Col } A$.
- The **rank** of A is the dimension of the column space of A and is denoted by $\text{rank } A$.

Problem

Find the rank of the matrix A and describe the column space efficiently.

$$A = \begin{array}{c} \vec{A}_1 \quad \vec{A}_2 \quad \vec{A}_3 \quad \vec{A}_4 \quad \vec{A}_5 \\ \begin{bmatrix} 1 & 2 & 1 & 3 & 2 \\ 1 & 3 & 6 & 0 & 2 \\ 3 & 7 & 8 & 6 & 6 \end{bmatrix} \end{array}$$

Let us find the RREF form of A :

$$\begin{bmatrix} 1 & 2 & 1 & 3 & 2 \\ 1 & 3 & 6 & 0 & 2 \\ 3 & 7 & 8 & 6 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & 3 & 2 \\ 0 & 1 & 5 & -3 & 0 \\ 0 & 1 & 5 & -3 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & 3 & 2 \\ 0 & 1 & 5 & -3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{array}{c} \vec{R}_1 \quad \vec{R}_2 \quad \vec{R}_3 \quad \vec{R}_4 \quad \vec{R}_5 \\ \begin{bmatrix} 1 & 0 & -9 & 9 & 2 \\ 0 & 1 & 5 & -3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

Note that $\text{Col } A = \text{Span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 7 \end{bmatrix} \right\}$, and hence $\text{rank } A = 2$.

Note that $\{\vec{R}_1, \vec{R}_2\}$ is a linearly independent set and $\text{Span}\{\vec{R}_1, \vec{R}_2, \dots, \vec{R}_5\} = \text{Span}\{\vec{R}_1, \vec{R}_2\}$.
Also $x_1 \vec{R}_1 + x_2 \vec{R}_2 = \vec{0}$ only has the trivial solution. Similar statements hold for columns of A .
 $\text{Span}\{\vec{A}_1, \vec{A}_2, \dots, \vec{A}_5\} = \text{Span}\{\vec{A}_1, \vec{A}_2\}$ and $x_1 \vec{A}_1 + x_2 \vec{A}_2 = \vec{0}$ only has the trivial solution.

The rank of A is equal to the number of pivots in A .

Null Space

Definition

Let A be an $m \times n$ matrix. The **null space** of A , or **kernel** of A is defined as $\text{Ker } A = \text{Nul } A = \{ \vec{x} \in \mathbb{R}^n : A\vec{x} = \vec{0} \}$.

Problem

Find $\text{Nul } A$ for $A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & 1 \\ 2 & 3 & 3 \end{bmatrix}$.

$$\begin{aligned} A\vec{x} &= \vec{0} \rightarrow \begin{bmatrix} 1 & 2 & 1 & | & 0 \\ 0 & -1 & 1 & | & 0 \\ 2 & 3 & 3 & | & 0 \end{bmatrix} \xrightarrow{-2r_1+r_3} \begin{bmatrix} 1 & 2 & 1 & | & 0 \\ 0 & -1 & 1 & | & 0 \\ 0 & -1 & 1 & | & 0 \end{bmatrix} \xrightarrow{-r_2+r_3} \begin{bmatrix} 1 & 2 & 1 & | & 0 \\ 0 & -1 & 1 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \\ &\xrightarrow{\substack{-r_2+r_1 \\ -r_1}} \begin{bmatrix} 1 & 3 & 0 & | & 0 \\ 0 & 1 & -1 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \quad \begin{aligned} x &= -3t \\ y &= t \\ z &= t \end{aligned} \end{aligned}$$

$$\vec{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = t \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix} \rightarrow \text{Nul } A = \text{Span} \left\{ \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix} \right\} \text{, which is a line through the origin}$$

Also, since the spanning set of $\text{Nul } A$ consists of a single vector, we have: $\dim \text{Nul } A = 1$.

Theorem

Let A be an $m \times n$ matrix. Then, $\text{rank}(A) + \dim \text{Nul } A = n$.

Example: Consider $A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & 1 \\ 2 & 3 & 3 \end{bmatrix}$. After row-reduction, we saw that

$A \xrightarrow{\text{reduced to}} \begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$ which has two leading 1's, hence $\text{rank } A = 2$.

Also, we saw that $\dim \text{Nul } A = 1$. Furthermore A is 3×3 which means $n = 3$.

$$\text{rank}(A) + \dim \text{Nul } A = n.$$

$$2 + 1 = 3$$

The Invertible Matrix Theorem (continued)

Theorem

Let A be an $n \times n$ matrix. The following conditions are equivalent.

- A is an invertible matrix.
- $\text{Col } A = \mathbb{R}^n$.
- $\text{rank } A = n$.
- $\text{Nul } A = \{\vec{0}\}$.

In the past, we discussed the following equivalences (in black), which we can express in terms of rank, null space and column space of A (in red):

A is an invertible matrix. $\iff$ The columns of A are linearly independent.
 $\text{rank } A = n$

A is an invertible matrix. $\iff$ The columns of A span $\mathbb{R}^n$.
 $\text{Col } A = \mathbb{R}^n$

A is an invertible matrix. $\iff$ $A\vec{X} = 0$ has only the trivial solution, $\vec{X} = 0$.
 $\text{Nul } A = \{\vec{0}\}$